

Benchmarking Class-Imbalance Mitigation Across Four Histopathology Datasets

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Abstract

Class imbalance in computational pathology depends on the input regime: labeled image patches support ordinary classification, whereas whole-slide image (WSI) diagnosis is commonly modeled as weakly supervised learning over patch-feature bags. This report benchmarks representative class-imbalance mitigation methods on four histopathology datasets—CAMELYON16, TCGA-UT, BRACS, and PANDA—presented in order of the diagnostic question they pose (detection, typing, subtyping, grading) and chosen to span the difficulty–imbalance plane while varying organ, label type, and annotation form. Imbalance is characterised primarily by the head-to-tail ratio $\rho = N_{\max}/N_{\min}$ —the standard imbalance factor of the long-tailed-recognition literature—with the entropy-based $1 - H_{\text{norm}} \in [0, 1]$ as a class-count-invariant cross-check, both reported separately at patch and slide level. All benchmarks fix a frozen Virchow2 encoder, a bounded per-slide evidence budget, and patient-disjoint splits. On CAMELYON16 (the suite’s highest native patch imbalance, $\rho \approx 39$; near-balanced slide labels, $\rho \approx 1.5$) every method reaches near-ceiling discrimination and the unweighted baseline is competitive. On TCGA-UT (mild imbalance, $\rho \approx 28$, 32-class typing) centre-focused affinity loss (CFAL) reaches the highest patch-level macro F1, OKO set learning the highest balanced accuracy and the best raw NLL and Brier among patch methods, while RankMix and MDE-MIL lead the WSI-bag regime. On BRACS (mild patch imbalance, $\rho \approx 4.6$, fine-grained subtyping) only OKO and CFAL separate from the CE baseline. On PANDA (mild imbalance, two targets of different difficulty on the same biopsies) the no-mitigation baseline is competitive in both regimes. Because these native comparisons confound task difficulty, training support, and imbalance, the controlled analysis fixes training-set size and evaluates imbalance-induced deficits against a balanced CE reference. Mitigation is then expressed as the fraction of that deficit recovered, with calibration and computational cost reported separately and independent feature-space probes used to test predictors of recovery.

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1 Introduction

Class imbalance changes what a pathology classifier is rewarded for learning. When common tumor types dominate training data, high aggregate accuracy can coexist with unreliable rare-class recall, especially if evaluation emphasizes prevalence-sensitive metrics. Long-tailed learning and medical-image evaluation literature therefore motivate classwise recall, macro F1, balanced accuracy, and calibration-aware probability metrics as primary outcomes in imbalanced settings (Saini and Susan, 2023; Zhang et al., 2023; Mosquera et al., 2024).

Imbalance methods often assume a specific prediction unit. Image-level methods use labeled images, image embeddings, or directly generated image samples, whereas WSI-level methods use weak labels over bags of instances together with attention, instance ranking, or bag-level representation learning. The computational-pathology pipeline these regimes sit in—whole-slide imaging, tiling into patches, and weakly supervised multiple-instance learning over frozen foundation-model features—and the full method definitions, imbalance mechanisms, and dataset provenance are documented in two companion reports (Queisler, 2026a,b); this report contributes the experimental results and cites them only where a definition is needed.

This benchmark is organised around three research questions, evaluated across four datasets presented in order of the diagnostic question they pose—CAMELYON16 detection, TCGA-UT typing, BRACS subtyping, and PANDA grading. Each question concerns a controlled contrast in which the task, patient split, frozen encoder, training-set size, and held-out evaluation set are fixed:

RQ1 — Effect of imbalance. How does increasing training imbalance affect tail-class discrimination and calibration in patch classification and WSI-level MIL?

RQ2 — Mitigation effectiveness. Which methods recover the performance loss caused by training imbalance? What calibration and computational costs accompany that recovery?

RQ3 — Predictors of mitigation headroom. Do independently measured class separability and effective sample size predict mitigation recovery better than the head-to-tail ratio, and does this relationship differ between patch and WSI supervision?

RQ1 first tests whether imbalance creates a measurable deficit. RQ2 then conditions mitigation claims on that deficit rather than treating any improvement over imbalanced CE as recovered performance. RQ3 tests candidate explanations measured independently of the mitigation runs, avoiding the circular use of CE test performance as a proxy for separability. The four datasets span distinct combinations of label granularity, sample support, and apparent feature-space overlap; CAMELYON16 additionally permits a pathology-specific MIL analysis in which between-slide class imbalance and within-slide informative-instance sparsity are manipulated separately.

2 Datasets

The four datasets are summarised in Table 1; this section states only what the benchmark consumes from each. Imbalance is quantified primarily by the head-to-tail ratio $\rho = N_{\max}/N_{\min}$ (the standard long-tailed-recognition imbalance factor), with the entropy-based $1 - H_{\text{norm}} \in [0, 1]$ as a class-count-invariant cross-check; both are reported separately at patch and slide level and defined in §2.1. All benchmarks use frozen Virchow2 2,560-dimensional features (concatenation of CLS token and mean patch token, cast to float32) and patient-disjoint train/validation/test splits (70/15/15) across three random seeds (Vorontsov et al., 2024; Zimmermann et al., 2024).

Dataset	Target label(s)	Split unit	Budget (med. tiles)	Ratio ρ		$1 - H_{\text{norm}}$	
				Patch	Slide	Patch	Slide
CAMELYON16	Tumor/normal (patch); metastasis (WSI)	Slide = patient	7874	39.2	1.5	0.832	0.028
TCGA-UT	32 cancer types	Case (patient)	30	28.1	28.3	0.085	0.085
BRACS	7 breast-lesion subtypes	Patient	360	4.6	2.3	0.072	0.018
PANDA	Cancer/benign (patch); ISUP 0–5 (WSI)	Biopsy	421	4.7	2.4	0.332	0.043

Table 1: Benchmark dataset overview, in the order the datasets are discussed. The evidence budget is the per-slide cap (data-driven median of available tiles) used in both regimes. Imbalance is reported primarily as the head-to-tail ratio $\rho = N_{\text{max}}/N_{\text{min}}$ (the long-tailed-recognition imbalance factor), with the entropy-based $1 - H_{\text{norm}}$ as a class-count-invariant cross-check; for PANDA the patch and slide figures describe two different targets (binary cancer/benign vs. six-class ISUP).

CAMELYON16. Binary labels are mask-derived: each tile is labelled tumor or normal by intersecting it with the published annotation mask. Of 399 slides, 398 have usable tile evidence; the 20 tumor slides whose annotations are not exhaustive are excluded from the patch regime but retained in the WSI-bag regime, where the reliable slide-level label is used. The per-slide evidence budget is the empirical median of available tissue tiles (7,874)—two orders of magnitude larger than TCGA-UT’s 30-tile cap, reflecting that a whole lymph-node section yields far more tissue than a single lesion ROI or tumor-region crop. Each slide is its own patient, so slide-disjoint and patient-disjoint splits coincide. The severe patch imbalance ($\rho \approx 39$, $1 - H_{\text{norm}} \approx 0.832$)—the highest in the suite—is a within-slide effect: tumor tiles are scarce even on slides that do contain a metastasis, so the slide-level labels are nearly balanced ($\rho \approx 1.5$, $1 - H_{\text{norm}} \approx 0.028$).

TCGA-UT. The dataset contains 1,608,060 patches from 8,736 diagnostic slides across 32 cancer types (Tomczak et al., 2015; Komura et al., 2022). Splits are assigned at the TCGA participant level (case-disjoint), using 5,025 training, 1,075 validation, and 1,075 test cases per seed; all slides from one participant remain in one partition. Because every patch inherits its slide’s cancer-type label, the patch and slide imbalance coincide ($\rho \approx 28$, $1 - H_{\text{norm}} \approx 0.085$). The patch benchmark caps each slide at 30 deterministic patches (empirical median of available tiles at resolution level 0), yielding a controlled manifest shared by all patch methods; the WSI-bag benchmark applies the same 30-instance cap to each slide’s feature bag.

BRACS. Evidence is derived from the ROI set: each annotated ROI is tiled into 256×256 RGB patches, and each slide’s pooled ROI tiles are capped at the empirical median (360 tiles per slide). This design preserves label fidelity because all instances originate from pathologist-selected lesion regions. Of 547 published WSIs, 387 contribute at least one ROI tile; the benchmark covers 151 patients and 99,352 extracted patches over the seven-subtype target, a mild imbalance ($\rho \approx 4.6$, $1 - H_{\text{norm}} \approx 0.072$ patch; $\rho \approx 2.3$, $1 - H_{\text{norm}} \approx 0.018$ slide).

PANDA. The WSI-bag regime uses the canonical six-class ISUP grade (0 = benign, 1–5 = increasing cancer grade groups; $\rho \approx 2.4$, $1 - H_{\text{norm}} \approx 0.043$). The patch regime uses a provider-consistent binary cancer/benign target derived from pixel masks ($\rho \approx 4.7$, $1 - H_{\text{norm}} \approx 0.332$); because Radboud and Karolinska masks use institution-dependent value semantics, collapsing both to cancer/benign is the only label the two providers share. To keep frozen-feature extraction tractable, the benchmark uses a stratified subset of approximately 2,000 biopsies, sampled

to preserve the joint distribution of ISUP grade and provider. The per-slide evidence budget is the empirical median number of tissue tiles per biopsy (421); biopsies without available masks are excluded from the patch regime but kept in the WSI-bag regime.

2.1 Quantifying imbalance

Imbalance is reported primarily by the head-to-tail ratio, the imbalance factor standard to the long-tailed-recognition literature the evaluated methods come from:

$$\rho = \frac{N_{\max}}{N_{\min}} \geq 1, \quad (1)$$

the largest-to-smallest class support, with $\rho = 1$ perfectly balanced and larger values more skewed. Cui et al. and Buda et al. parameterise their long-tailed benchmarks directly by ρ (imbalance factors up to 200), so reporting ρ places each dataset on the same axis as the methods’ design regime and as the controlled construction of §3.1 (Cui et al., 2019; Buda et al., 2018).

The ratio has one blind spot: it depends only on the two extreme classes and ignores the shape of the distribution between them, and it is not comparable across datasets with different class counts. As a class-count-invariant cross-check, an entropy-based measure is therefore also reported. For empirical class proportions $p_1, \dots, p_K$ (with $\sum_i p_i = 1$), the Shannon entropy $H(p) = -\sum_{i=1}^K p_i \log p_i$ attains its maximum $\log K$ at the uniform distribution; normalising and complementing gives

$$1 - H_{\text{norm}} = 1 - \frac{-\sum_{i=1}^K p_i \log p_i}{\log K} \in [0, 1], \quad (2)$$

with 0 perfectly balanced and values toward 1 more skewed. The $\log K$ normalisation removes the dependence on the number of classes, so $1 - H_{\text{norm}}$ compares a two-class cohort against a thirty-two-class one on a common scale where ρ cannot. Both are reported separately at patch and slide level, because the two levels can differ sharply when patch labels are derived from masks rather than inherited from the slide; the full native class distributions are given in Queisler (2026a).

The two measures answer different questions. Because $1 - H_{\text{norm}}$ is a $\log K$ -normalised average deviation from uniformity, a fixed ρ maps to very different $1 - H_{\text{norm}}$ depending on K : CAMELYON16 patches ($\rho \approx 39$, $K = 2$) score $1 - H_{\text{norm}} \approx 0.832$, whereas TCGA-UT patches at a comparable ratio ($\rho \approx 28$, $K = 32$) score only $1 - H_{\text{norm}} \approx 0.085$. The reverse also holds: an artificially long-tailed distribution at imbalance factor $\rho = 100$ —a regime everyone agrees is severe—still scores a low $1 - H_{\text{norm}} \approx 0.13$ once it has a hundred classes, because entropy is then dominated by the bulk of moderately populated classes and is nearly insensitive to how extreme the single rarest class is. Neither behaviour is a defect: ρ measures the extremity of the tail while $1 - H_{\text{norm}}$ measures how far the *whole* distribution sits from uniform, and the two coincide only at $K = 2$. Reading them together— ρ as the headline, $1 - H_{\text{norm}}$ as the K -invariant cross-check—avoids being misled by either the tail-blindness of the entropy measure or the class-count-blindness of the ratio.

Native imbalance is mild relative to the regime these methods target. At *slide* level—the unit every diagnostic target in this benchmark is ultimately scored on—all four datasets are only mildly skewed, whether read as $1 - H_{\text{norm}} \in [0.018, 0.085]$ or as a head-to-tail ratio $\rho \leq 28.3$ (Table 1). The two large $1 - H_{\text{norm}}$ values in the table (CAMELYON16 patch 0.832, PANDA patch 0.332) are within-slide, binary mask effects over modest underlying ratios ($\rho \approx 39$ and 4.7), not properties of the typing, subtyping, or grading targets themselves.

This matters because the long-tailed-recognition methods evaluated here were designed for, and validated at, far harsher imbalance: Cui et al. construct long-tailed benchmarks by exponential class-count decay at imbalance factors $\rho = N_{\max}/N_{\min}$ up to 200 (Cui et al., 2019), and Buda et al. study step and linear imbalance at comparable ratios (Buda et al., 2018). Figure 1 places the native ratios against this $\rho \in [100, 200]$ stress-test band: every slide-level target sits an order of magnitude below it. Benchmarking imbalance-mitigation methods on the native distributions therefore under-tests them, which motivates the controlled construction of §3.1.

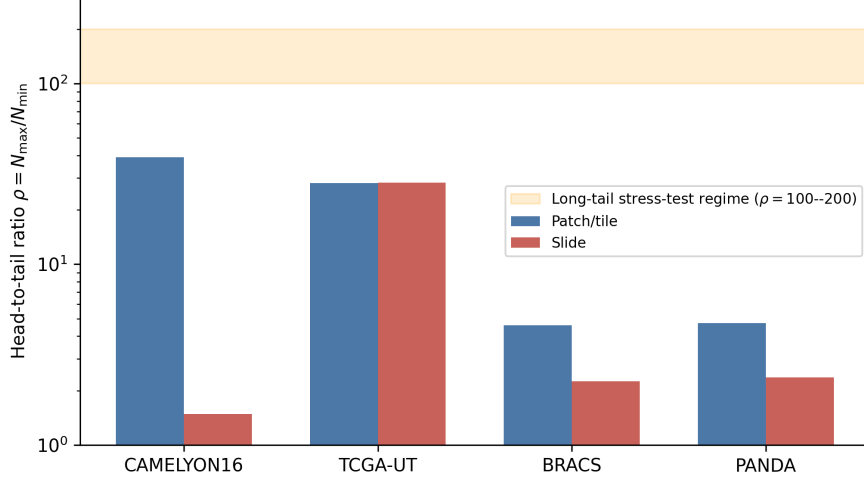


Figure 1: Native head-to-tail ratio $\rho = N_{\max}/N_{\min}$ at patch/tile and slide level for all four datasets (log scale), against the $\rho \in [100, 200]$ regime at which the evaluated long-tailed-recognition methods were designed and validated (Cui et al., 2019; Buda et al., 2018). Every slide-level target—the unit each diagnostic question is scored on—sits an order of magnitude below the band.

3 Experimental Setup

The methods, training protocol, and tuning grids below are shared across all four datasets; full method definitions and metric formulas follow Queisler (2026b). Table 2 lists the method roster by family. CE and MIL CE are unweighted cross-entropy baselines, included so that mitigation gains are measured against a genuine no-mitigation reference; patch and WSI-bag methods are never ranked against each other.

Family	Methods	Regime
No-mitigation baseline	CE, MIL CE	Patch, WSI bag
Data-level sampling and augmentation	Balanced sampling, RankMix	Patch, WSI bag
Algorithm-level losses	Weighted CE, Focal loss, CFAL	Patch (CFAL); Patch and WSI bag (others)
Hybrid sampling-loss	CE + soft F1, CE + soft MCC, OKO, MDE-MIL	Patch (first three, OKO); WSI bag (MDE-MIL)
Synthetic data generation	ProGAN augmentation	Patch
Representation and architecture	SC-MIL	WSI bag

Table 2: Method families and regimes. Method definitions follow Queisler (2026b).

Both benchmarks share a fixed training protocol (Table 9): AdamW at learning rate 10^{-3} , weight decay 10^{-4} , 30 epochs with no early stopping, final-epoch model selection, and seeds

0–2. The only per-method degree of freedom is a single imbalance-specific control, swept on validation using the grids in Table 10; identical grids are applied on all four datasets. ProGAN sweeps only the final-depth refinement epochs, with generator training following Table 11. These three protocol tables are collected in Appendix A.

The patch and WSI-bag classifier heads (Figure 2) are shared across all datasets. The patch head is a two-layer MLP (linear map to 512 units, ReLU, dropout 0.1, linear output to C classes); CFAL keeps the trunk but replaces the final linear map with one learnable prototype per class and Gaussian affinity-based prediction. The WSI-bag head is an attention-based MIL network: each instance passes through a shared encoder ($2560 \rightarrow 256$), instance weights come from a linear scorer with softmax over instances, the bag representation is their weighted sum, and slide logits come from a final linear map. SC-MIL adds a two-layer projection head of width 256; MDE-MIL keeps the attention pooling but uses two expert heads. The WSI-bag input is capped at the dataset-specific evidence budget (Table 1).

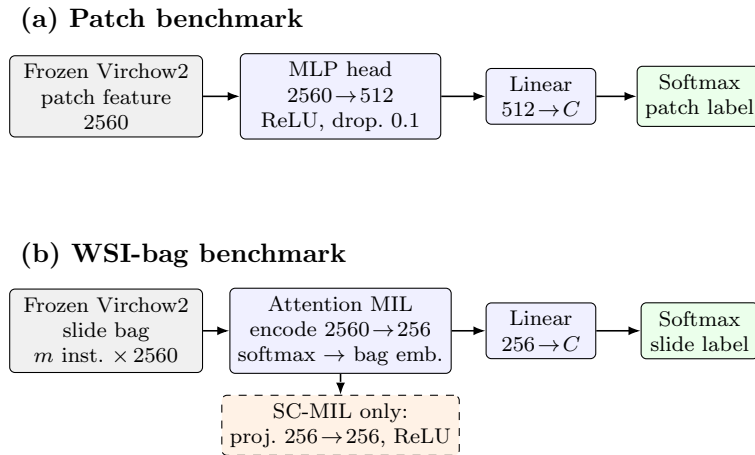


Figure 2: Shared classifier heads for the benchmark (C = number of classes in each dataset). Panel (a) classifies one controlled patch embedding per forward pass; CFAL reuses the MLP trunk but replaces the linear $512 \rightarrow C$ head with learnable prototypes and affinity-based prediction. Panel (b) encodes all instances on a slide, pools them with attention, and predicts one slide-level label; the dashed branch is used only by SC-MIL, and MDE-MIL replaces the single linear head with two expert heads. Virchow2 features are frozen in both benchmarks.

Macro F1 is the primary ranking metric within each regime because it weights classes equally and penalizes failure on rare classes; balanced accuracy and accuracy are reported alongside it. Three complementary probability-quality metrics—negative log likelihood (NLL), multiclass Brier score, and expected calibration error (ECE)—are reported on held-out softmax outputs, both before and after a unified post-hoc temperature-scaling protocol in which, for each method and seed, a single temperature is fit on validation by minimizing NLL and applied unchanged to the test scores. Full metric definitions and the temperature-scaling protocol are given in Queisler (2026b).

RQ1 gives priority to tail-class recall and tail-class one-versus-rest AUROC alongside macro F1 and balanced accuracy, because aggregate gains can conceal deterioration in the rare classes that motivate the intervention. RQ2 reports calibration as a separate outcome rather than folding it into discrimination, and measures computational cost by training wall time, accelerator-hours, and peak accelerator memory relative to the corresponding CE run under the same training budget.

All four datasets use three random seeds, deterministic per-slide evidence caps, and frozen Virchow2 features. Hyperparameters are selected independently per method: for each candidate configuration the validation macro F1 is averaged over the three seeds and the configuration

with the highest mean is chosen (balanced accuracy as tie-breaker); a single configuration per method is then fixed before evaluating on the held-out test split. Results are reported as mean $\pm$ standard deviation across seeds, and patch and WSI-bag methods are compared only within their own regime.

3.1 Controlled imbalance construction

Section 2.1 showed that every native target is only mildly imbalanced relative to the long-tail stress-test regime the evaluated mitigation methods were designed for. Rather than accepting that mismatch, each dataset can be restructured to a controlled, artificially skewed distribution, applied *independently* at patch and slide level to the class distribution each regime is scored on. Three constructions from the long-tailed-recognition literature are used—truncated exponential decay, step imbalance, and (for binary targets) ratio downsampling—each enforcing a hard minimum per-class support N_{floor} so that no class is starved below what stable evaluation requires. The floor is grounded in the sampling precision of per-class recall. Under a binomial approximation for recall estimated from n test examples, the standard error satisfies

$$\text{SE}(\hat{R}) = \sqrt{\frac{R(1-R)}{n}} \leq \frac{1}{2\sqrt{n}}, \quad N_{\text{floor}}(\varepsilon) = \frac{1}{4\varepsilon^2 f_{\text{test}}}, \quad (3)$$

where $f_{\text{test}} = 0.15$ and $n = f_{\text{test}} N_{\text{floor}}$. Thus the patch floor $N_{\text{floor}} = 200$ yields approximately thirty test patches per class and a worst-case $\text{SE} \leq 0.091$. The analogous precision target is infeasible for slides: $N_{\text{floor}} = 20$ yields approximately three test slides and the coarser bound $\text{SE} \leq 0.289$, while remaining below the rarest retained TCGA-UT type (DLBC, 28 slides). The two floors therefore follow the same standard-error formula but reflect the different attainable precision at patch and slide level. The constructions only ever downsample.

For the causal contrasts in RQ1–RQ3, these target distributions serve as severity envelopes rather than as unrelated training sets. Within each dataset and regime, a balanced CE condition and every imbalanced condition use the same total training support T , the same patient split, and the same unchanged validation and test sets. The largest feasible shared T is chosen subject to the native per-class support constraints at every compared severity; only the training allocation across classes changes. Samples are nested where possible so that increasing imbalance removes examples from the same common pool instead of changing case composition arbitrarily. Encoder, evidence budget, optimisation schedule, and evaluation set remain fixed. Consequently, differences between balanced and imbalanced CE estimate the effect of training imbalance rather than a mixture of imbalance and training-set size.

Truncated exponential decay (Cui et al.). Ranking classes by native support, the target count for the i -th ranked class ($i = 0, \dots, K - 1$) is

$$n_i = \max(N_{\text{floor}}, \lfloor N_{\text{max}} \mu^i \rfloor), \quad \mu \in (0, 1), \quad (4)$$

with μ chosen to hit a target imbalance factor $\rho = N_{\text{max}}/N_{\text{min}}$ (Cui et al., 2019). The floor caps the decay so that every class retains enough examples for macro-averaged evaluation instead of thinning to a handful.

Step imbalance (Buda et al.). Classes are split into a head group, which keeps its native support, and a tail group, which is uniformly downsampled to N_{floor} (Buda et al., 2018). This isolates rarity as a binary property (in the head group or not) rather than a continuous rank effect, and lets the tail group be chosen for clinical relevance rather than by native rank. For binary targets neither scheme applies, so the minority class is simply ratio-downsampled to a target ρ , floored. For the multi-class exponential and step constructions, the achieved ratio is not an additional free parameter: once the head retains its native support, it follows as

$\rho_{\text{target}} = N_{\text{max}}^{\text{native}}/N_{\text{floor}}$. Binary patch targets instead need an external anchor because their millions of majority patches would make that rule degenerate. They use $\rho = 100$, the entry of the $\rho \in [100, 200]$ regime in which the evaluated methods were validated (Cui et al., 2019; Buda et al., 2018).

Per-dataset strategy. Table 3 lists the construction at both levels, in the order of Table 1:

- **CAMELYON16** (binary at both levels). Slides are ratio-downsampled to a 10:1 normal:tumor prevalence (a low-prevalence screening scenario), which the small cohort caps near $\rho \approx 10$; tumor patches, being abundant, are ratio-downsampled all the way to the stress regime ($\rho = 100$).
- **TCGA-UT** (32 types, shared by both levels) uses exponential decay ranked by support. The slide target widens to $\rho \approx 40$, but the patch target barely moves ($\rho \approx 28$): the rarest type (DLBC) has only 168 patches, so the patch floor binds before the decay can steepen—an honest ceiling on how imbalanced the patch regime can be made without starving a class below evaluability.
- **BRACS** (7 subtypes, shared) uses step imbalance: the benign/usual head {N, PB, UDH, IC} keeps native support while the clinically critical atypical/malignant tail {FEA, ADH, DCIS} is downsampled to the floor, reaching $\rho \approx 10$ (slide) and $\rho \approx 149$ (patch).
- **PANDA** (6 ISUP grades on slides, binary cancer/benign on patches). The slide target uses exponential decay fixed to the ordinal grade order (respecting grade order) on the full 10,616-slide training cohort driving ρ to ≈ 145 ; patches are ratio-downsampled to the stress regime ($\rho = 100$).

Figure 3 shows the resulting target ratios against the same stress-test band as the native Figure 1: the construction lifts every dataset-level pair into or toward the regime the methods were designed for, the sole exception being the TCGA-UT patch target noted above. Table 3 lists the target support, ratio, and $1 - H_{\text{norm}}$ at both levels, and Appendix B plots the full native-versus-target distributions. This construction is proposed as the basis for a future benchmark iteration; the results in §4 use the native distributions.

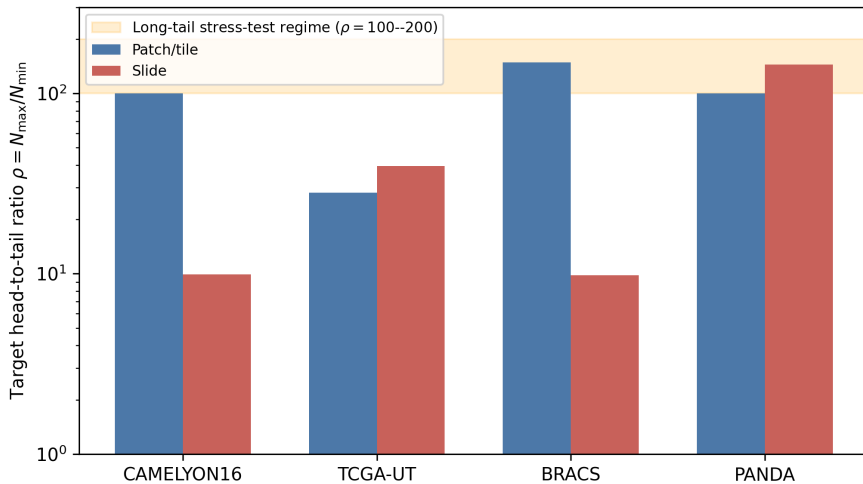


Figure 3: Target head-to-tail ratio ρ at patch and slide level after the controlled construction, against the $\rho \in [100, 200]$ stress-test band—the target counterpart of the native Figure 1. The construction lifts every dataset-level pair into or toward the regime the evaluated methods were designed for, except the TCGA-UT patch target, whose rarest type is patch-scarce.

Dataset	Level	K	Method	Support	ρ	$1 - H_{\text{norm}}$
CAMELYON16	Patch	2	Ratio (100:1)	2,399,712 $\rightarrow$ 2,363,440 (98%)	39.2:1 $\rightarrow$ 100.0:1	0.832 $\rightarrow$ 0.920
	Slide	2	Ratio (10:1)	398 $\rightarrow$ 262 (66%)	1.5:1 $\rightarrow$ 9.9:1	0.028 $\rightarrow$ 0.558
TCGA-UT	Patch	32	Exp. decay	52,194 $\rightarrow$ 45,723 (88%)	28.1:1 $\rightarrow$ 28.1:1	0.085 $\rightarrow$ 0.100
	Slide	32	Exp. decay	8,736 $\rightarrow$ 6,837 (78%)	28.3:1 $\rightarrow$ 39.6:1	0.085 $\rightarrow$ 0.126
BRACS	Patch	7	Step	99,352 $\rightarrow$ 73,047 (74%)	4.6:1 $\rightarrow$ 148.7:1	0.072 $\rightarrow$ 0.317
	Slide	7	Step	851 $\rightarrow$ 625 (73%)	2.3:1 $\rightarrow$ 9.8:1	0.018 $\rightarrow$ 0.151
PANDA	Patch	<i>Full-cohort supports and targets pending computation</i>				
	Slide	<i>Full-cohort supports and targets pending computation</i>				

Table 3: Per-dataset controlled-imbalance construction at patch and slide level, in the order of Table 1. Each of the Support, ρ , and $1 - H_{\text{norm}}$ columns reads native $\rightarrow$ target; Support is the total class-summed count (patches or slides), with the percentage of the native total retained after downsampling. PANDA values are withheld pending full-cohort computation on all 10,616 slides; the currently available values use only a 2,000-slide feature-extraction subset and are not reported as full-cohort results.

For a discrimination metric M oriented so that larger values are better, the imbalance-induced deficit and the recovery fraction are defined as

$$\begin{aligned}
D_M &= M(\text{balanced CE}) - M(\text{imbalanced CE}), \\
R_M &= \frac{M(\text{method, imbalanced}) - M(\text{imbalanced CE})}{D_M}.
\end{aligned} \tag{5}$$

Here $R_M = 0$ denotes no recovery, $R_M = 1$ denotes recovery to the balanced-CE reference, $R_M < 0$ denotes further deterioration, and $R_M > 1$ denotes improvement beyond that reference. Recovery is reported only when D_M is positive and distinguishable from seed variation; otherwise the condition provides no evidenced imbalance deficit to recover. NLL, Brier score, and ECE retain their natural lower-is-better orientation and are reported as paired changes, avoiding an opaque composite of discrimination and calibration.

RQ3 uses predictors computed before any mitigation run. Feature-space separability is summarised by balanced k -nearest-neighbour macro recall, a class-balanced linear probe, and per-class nearest-neighbour error on frozen Virchow2 representations. Patch analyses operate on patch embeddings; WSI analyses use a fixed, non-learned mean of the capped patch embeddings so that the probe does not inherit the CE or MIL model being explained. Effective sample size is estimated per class with the standard cluster correction

$$N_{\text{eff},c} = \frac{N_c}{1 + (\bar{m}_c - 1) \text{ICC}_c}, \tag{6}$$

where patches are clustered within patients/slides and multiple slides, where present, are clustered within patients. Predictive analyses compare separability, N_{eff} , and ρ against R_M without using CE test performance as a separability covariate.

CAMELYON16 provides a prespecified MIL-specific secondary analysis motivated by the distinction between slide-level and instance-level imbalance in SC-MIL (Juyal et al., 2024). The number of positive slides and the fraction of mask-positive instances retained within each tumor bag are varied factorially while total slide support and bag size are fixed. This separates between-slide class imbalance from within-slide informative-instance sparsity and tests their interaction on WSI-level discrimination and calibration.

4 Results

The four datasets are reported in the order of Table 1. Each dataset carries one results float stacking the patch-feature and WSI-bag regimes, followed by three short paragraphs on patch-level results, bag-level results, and calibration. Absolute scores and method orderings are never comparable across the two regimes. TCGA-UT receives the deepest analysis; its tuning-sensitivity, paired-seed, and classwise-difficulty diagnostics, and all raw calibration tables, are collected in Appendices C and D.

4.1 CAMELYON16 — metastasis detection

CAMELYON16 is the most imbalanced dataset at patch level ($\rho \approx 39$, $1 - H_{\text{norm}} \approx 0.832$) yet nearly balanced at slide level ($\rho \approx 1.5$, $1 - H_{\text{norm}} \approx 0.028$), and its detection task is the most separable in the suite. The same frozen features separate tumor from normal tissue almost perfectly, so despite the severe patch imbalance every method reaches near-ceiling discrimination in both regimes (Table 4).

<i>Patch-feature regime</i>			
Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.997 ± 0.002	0.961 ± 0.015	0.966 ± 0.008
Balanced sampling	0.997 ± 0.001	0.969 ± 0.014	0.960 ± 0.020
ProGAN augmentation	0.996 ± 0.001	0.956 ± 0.009	0.959 ± 0.019
CE	0.996 ± 0.001	0.962 ± 0.015	0.958 ± 0.020
Focal loss	0.996 ± 0.001	0.972 ± 0.010	0.957 ± 0.027
OKO	0.996 ± 0.000	0.974 ± 0.007	0.955 ± 0.034
CE + soft F1	0.996 ± 0.001	0.986 ± 0.002	0.952 ± 0.035
Weighted CE	0.996 ± 0.002	0.966 ± 0.012	0.952 ± 0.028
CE + soft MCC	0.995 ± 0.001	0.982 ± 0.006	0.940 ± 0.050
<i>WSI-bag regime (slide-level metastasis detection)</i>			
Method	Accuracy	Balanced accuracy	Macro F1
MDE-MIL	0.983 ± 0.017	0.984 ± 0.018	0.982 ± 0.018
SC-MIL	0.983 ± 0.017	0.984 ± 0.018	0.982 ± 0.018
Balanced sampling	0.977 ± 0.010	0.979 ± 0.012	0.977 ± 0.010
Focal loss	0.977 ± 0.010	0.979 ± 0.012	0.977 ± 0.010
RankMix	0.972 ± 0.020	0.974 ± 0.020	0.971 ± 0.020
MIL CE	0.966 ± 0.017	0.965 ± 0.021	0.965 ± 0.018
Weighted CE	0.966 ± 0.017	0.965 ± 0.021	0.965 ± 0.018

Table 4: CAMELYON16 native test results for validation-selected configurations, mean $\pm$ standard deviation across seeds.

In the patch regime, balanced accuracy ranges from 0.956 to 0.986 and the unweighted CE baseline (BAcc 0.962) sits squarely inside the field. In the WSI-bag regime, all methods score between balanced accuracy 0.965 and 0.984; the unweighted MIL CE baseline is again fully competitive (BAcc 0.965), and the entire field is separated by less than 0.02 macro F1—canonical slide-level metastasis detection is nearly solved from bounded Virchow2 feature bags. Calibration follows discrimination (Appendix D): raw patch ECE is 0.003–0.005 for every method except

CFAL, and temperature scaling brings it to 0.001–0.002, markedly better than BRACS—the calibration counterpart of the separability gap.

4.2 TCGA-UT — pan-cancer typing

TCGA-UT widens the task to 32-class pan-cancer typing at a mild imbalance ($\rho \approx 28$, $1 - H_{\text{norm}} \approx 0.085$, identical at patch and slide level because every patch inherits its slide label). It receives the study’s deepest analysis: how much the selected control matters varies sharply across methods—across-configuration variation exceeds twice the seed standard deviation only for OKO and RankMix, while every single-knob resampling, reweighting, and focal mechanism produces variation within seed noise (Appendix C).

<i>Patch-feature regime</i>			
Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.847 ± 0.008	0.789 ± 0.007	0.798 ± 0.009
OKO	0.841 ± 0.006	0.791 ± 0.005	0.788 ± 0.003
CE + soft F1 (balanced)	0.835 ± 0.004	0.781 ± 0.002	0.780 ± 0.003
Weighted CE	0.826 ± 0.008	0.767 ± 0.006	0.774 ± 0.005
CE	0.828 ± 0.010	0.764 ± 0.011	0.773 ± 0.015
Balanced sampler	0.827 ± 0.010	0.769 ± 0.007	0.771 ± 0.014
CE + soft MCC (balanced)	0.824 ± 0.009	0.772 ± 0.004	0.770 ± 0.005
ProGAN augmentation	0.824 ± 0.004	0.760 ± 0.005	0.766 ± 0.002
Focal	0.825 ± 0.007	0.758 ± 0.007	0.765 ± 0.009

<i>WSI-bag regime</i>			
Method	Accuracy	Balanced accuracy	Macro F1
MDE-MIL (ensemble)	0.896 ± 0.002	0.849 ± 0.014	0.865 ± 0.011
RankMix	0.890 ± 0.005	0.864 ± 0.002	0.854 ± 0.008
SC-MIL	0.883 ± 0.013	0.837 ± 0.013	0.839 ± 0.008
Weighted MIL	0.882 ± 0.006	0.833 ± 0.019	0.835 ± 0.011
Balanced MIL	0.875 ± 0.003	0.829 ± 0.010	0.831 ± 0.009
MIL CE	0.873 ± 0.013	0.823 ± 0.014	0.826 ± 0.013
Focal MIL	0.872 ± 0.006	0.823 ± 0.019	0.824 ± 0.008

Table 5: TCGA-UT validation-selected test results over three seeds. Methods share the frozen Virchow2 encoder, the shared head family, and the evaluation code within each regime.

The patch benchmark compresses into a narrow performance band once patches are represented with frozen Virchow2 embeddings. CFAL ranks highest on macro F1 and OKO highest on balanced accuracy, both with paired gains over plain CE in all three seeds (Appendix C); weighted CE, balanced sampling, and focal loss remain within the CE-centered band. ProGAN augmentation stays within the non-generative spread despite adding a large synthetic tail-class volume, with per-class Inception FID values comparable to a source study where GAN augmentation helped an end-to-end classifier (Ruiz-Casado et al., 2024); the frozen encoder—not generator quality—is the proposed explanation for the null downstream effect (Appendix C).

On WSI bags, RankMix has the highest mean balanced accuracy and MDE-MIL the highest mean macro F1, both above plain MIL CE on both headline metrics in every seed. Other MIL variants cluster closely below this pair, and focal MIL is unstable across seeds (competitive on seeds 0 and 2, collapsing on seed 1), giving the widest seed-to-seed standard deviation in the WSI benchmark. On raw probability quality, CFAL improves discrimination most but is the most over-confident patch method (highest raw ECE, worst NLL/Brier), while OKO is the exception to the usual discrimination–calibration trade-off, attaining the best raw NLL and Brier among patch methods; focal loss attains the smallest raw ECE. On WSI bags, MDE-MIL

and RankMix reach the lowest raw NLL, Brier, and ECE. After temperature scaling, CFAL reaches the best patch NLL and Brier and focal loss the lowest patch ECE, while the simpler MIL baselines reach lower post-hoc ECE than MDE-MIL or RankMix, so the raw WSI ECE advantage should be read partly as a confidence-scale mismatch (Appendix D).

4.3 BRACS — breast-lesion subtyping

BRACS returns to a single organ but sharpens to fine-grained seven-subtype breast-lesion classification at a mild imbalance ($\rho \approx 4.6$, $1 - H_{\text{norm}} \approx 0.072$ patch; $\rho \approx 2.3$, $1 - H_{\text{norm}} \approx 0.018$ slide). Absolute scores are far lower than on TCGA-UT because the task is intrinsically harder and the breast-only training pool is smaller; even task-specific BRACS models report only mid-60% weighted F1 (Pati et al., 2022; Stegmüller et al., 2023), so a frozen-feature benchmark under a bounded budget should not approach TCGA-UT-level discrimination.

<i>Patch-feature regime</i>			
Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.589 ± 0.027	0.497 ± 0.042	0.490 ± 0.038
OKO	0.565 ± 0.054	0.502 ± 0.052	0.488 ± 0.049
CE + soft F1	0.547 ± 0.034	0.480 ± 0.036	0.468 ± 0.033
CE + soft MCC	0.545 ± 0.043	0.472 ± 0.047	0.463 ± 0.044
Balanced sampling	0.552 ± 0.045	0.470 ± 0.043	0.461 ± 0.045
ProGAN augmentation	0.549 ± 0.049	0.460 ± 0.045	0.457 ± 0.049
Focal loss	0.533 ± 0.041	0.460 ± 0.037	0.452 ± 0.036
CE	0.548 ± 0.045	0.451 ± 0.056	0.451 ± 0.055
Weighted CE	0.531 ± 0.045	0.461 ± 0.040	0.450 ± 0.042
<i>WSI-bag regime</i>			
Method	Accuracy	Balanced accuracy	Macro F1
MIL CE	0.672 ± 0.020	0.524 ± 0.103	0.508 ± 0.045
Weighted CE	0.667 ± 0.029	0.510 ± 0.093	0.501 ± 0.055
Focal loss	0.691 ± 0.029	0.490 ± 0.032	0.498 ± 0.033
MDE-MIL	0.635 ± 0.033	0.489 ± 0.093	0.480 ± 0.051
Balanced sampling	0.635 ± 0.022	0.449 ± 0.022	0.462 ± 0.023
RankMix	0.603 ± 0.104	0.437 ± 0.087	0.446 ± 0.075
SC-MIL	0.539 ± 0.092	0.367 ± 0.083	0.383 ± 0.086

Table 6: BRACS native test results for validation-selected configurations, mean $\pm$ standard deviation across patient-disjoint seeds.

In the patch regime, OKO and CFAL are the only methods that visibly separate from the unweighted CE baseline, leading by roughly five balanced-accuracy points, while every other method—including ProGAN augmentation—falls inside one standard deviation of CE. In the WSI-bag regime, the unweighted MIL CE baseline has the highest balanced accuracy and macro F1 and no mitigation method improves on it—exactly the failure mode the no-mitigation baseline is included to expose. Calibration is severe: raw patch ECE is 0.29–0.42 for most methods, with OKO the exception (raw ECE 0.045, fitted temperature $T \approx 0.88$) and CFAL showing the near-degenerate low-temperature signature ($T \approx 0.08$) that scaling nonetheless corrects; WSI-bag calibration is uniformly worse (post-scaling ECE 0.136–0.168),

reflecting the smaller training pool and fine-grained label set (Appendix D).

4.4 PANDA — prostate cancer grading

PANDA poses two mildly imbalanced targets on the same biopsies at opposite ends of the difficulty axis: binary cancer/benign tile detection ($\rho \approx 4.7$, $1 - H_{\text{norm}} \approx 0.332$, patch regime) and six-class ordinal ISUP grading ($\rho \approx 2.4$, $1 - H_{\text{norm}} \approx 0.043$, WSI-bag regime), the latter a fine-grained ordinal problem on which even purpose-built challenge models reach only moderate agreement (Bulten et al., 2022).

<i>Patch-feature regime (cancer vs. benign tiles)</i>			
Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.928 ± 0.004	0.832 ± 0.003	0.866 ± 0.002
CE	0.921 ± 0.004	0.838 ± 0.003	0.859 ± 0.001
ProGAN augmentation	0.918 ± 0.005	0.839 ± 0.000	0.855 ± 0.002
Weighted CE	0.915 ± 0.006	0.845 ± 0.004	0.854 ± 0.003
Focal loss	0.896 ± 0.014	0.854 ± 0.006	0.834 ± 0.012
OKO	0.892 ± 0.004	0.860 ± 0.006	0.832 ± 0.006
Balanced sampling	0.890 ± 0.004	0.848 ± 0.003	0.826 ± 0.002
CE + soft F1	0.876 ± 0.009	0.852 ± 0.006	0.813 ± 0.005
CE + soft MCC	0.874 ± 0.009	0.852 ± 0.006	0.811 ± 0.004
<i>WSI-bag regime (six-class ISUP grading)</i>			
Method	Accuracy	Balanced accuracy	Macro F1
Balanced sampling	0.653 ± 0.028	0.588 ± 0.032	0.590 ± 0.033
Focal loss	0.654 ± 0.003	0.584 ± 0.016	0.583 ± 0.018
MDE-MIL	0.652 ± 0.029	0.581 ± 0.036	0.581 ± 0.040
RankMix	0.626 ± 0.014	0.571 ± 0.031	0.566 ± 0.034
MIL CE	0.639 ± 0.017	0.566 ± 0.016	0.565 ± 0.021
Weighted CE	0.639 ± 0.017	0.566 ± 0.016	0.565 ± 0.021
SC-MIL	0.619 ± 0.005	0.565 ± 0.025	0.559 ± 0.021

Table 7: PANDA native test results for validation-selected configurations, mean $\pm$ standard deviation across seeds. The two regimes carry different targets (binary detection vs. six-class ISUP), so their scores are not comparable.

In the patch regime, every method reaches balanced accuracy between 0.832 and 0.860 and macro F1 between 0.811 and 0.866; the unweighted CE baseline sits squarely inside this band (BAcc 0.838, macro F1 0.859) and no mitigation method separates from it by more than between-seed noise. The WSI-bag regime inverts the difficulty: macro F1 spans only 0.559–0.590—roughly 0.27 below the detection regime on the very same biopsies—and the unweighted MIL CE baseline (BAcc 0.566, macro F1 0.565) is within noise of the field; balanced sampling, focal loss, and MDE-MIL lead it by about 0.02, but between-seed standard deviations of 0.02–0.04 make these numerical rather than statistically separated leads. Calibration follows separability: patch raw ECE is 0.024–0.103 and post-scaling ECE 0.005–0.036 (OKO best), whereas the harder WSI-bag regime has raw ECE 0.129–0.252 and residual post-scaling ECE 0.039–0.097 (Appendix D).

4.5 Cross-Dataset Findings

Dataset	Regime	Best mitigation method	Baseline macro F1	Δ macro F1
CAMELYON16	Patch	CFAL	0.958	+0.008
CAMELYON16	WSI bag	MDE-MIL / SC-MIL	0.965	+0.017
TCGA-UT	Patch	CFAL	0.870	+0.008
TCGA-UT	WSI bag	MDE-MIL	0.843	+0.014
BRACS	Patch	CFAL	0.451	+0.039
BRACS	WSI bag	MIL CE (baseline)	0.508	0.000
PANDA	Patch	CFAL	0.859	+0.007
PANDA	WSI bag	Balanced sampling	0.565	+0.025

Table 8: Descriptive mitigation gains on the native training distributions. Baseline is CE (patch) or MIL CE (WSI bag). Δ macro F1 is the best mitigation method’s mean test macro F1 minus the no-mitigation baseline; it is not the recovery fraction in Equation 5, because the native runs have no size-matched balanced-CE reference. The BRACS WSI-bag row is 0.000 because no mitigation method beat MIL CE.

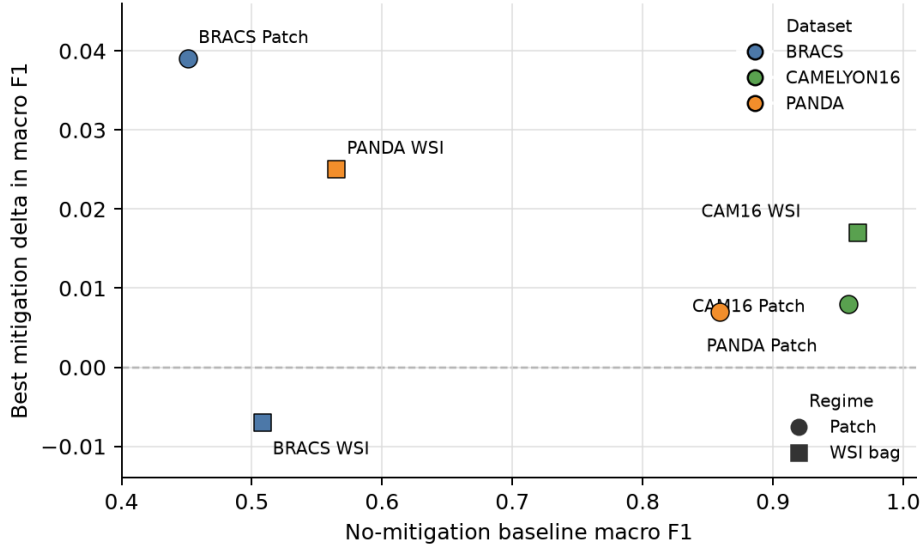


Figure 4: No-mitigation baseline macro F1 versus the descriptive native-distribution gain across all dataset–regime pairs. The mechanical association between a lower baseline and a larger possible gain is why this quantity is not used as mitigation recovery in RQ2.

Table 8 and Figure 4 summarise the completed native-distribution runs. They describe method behaviour but cannot establish the effect of imbalance or the fraction of imbalance-induced damage recovered: neither the training-set size nor a balanced-CE counterfactual is held fixed. Three patterns motivate, rather than answer, the controlled research questions.

Native imbalance severity does not rank the observed gains. CAMELYON16 has the highest patch-level imbalance in the suite ($\rho \approx 39$, $1 - H_{\text{norm}} \approx 0.832$) yet shows among the smallest mitigation deltas (+0.008 macro F1 at patch level). The largest positive deltas instead occur for BRACS patch ($\rho \approx 4.6$, $1 - H_{\text{norm}} \approx 0.072$, $\Delta = +0.039$) and PANDA WSI-bag ISUP grading ($\rho \approx 2.4$, $1 - H_{\text{norm}} \approx 0.043$, $\Delta = +0.025$). Because task, support, and separability also differ across these datasets, this comparison is hypothesis-generating rather than a test of which property causes mitigation headroom.

Patch- and slide-level imbalance describe different mechanisms. CAMELYON16’s severe patch imbalance is largely a within-slide scarcity of tumor instances, whereas its slide labels

are nearly balanced ($1 - H_{\text{norm}} \approx 0.028$). A single ratio therefore cannot represent both rare positive bags and sparse informative instances; the factorial CAMELYON16 analysis separates these mechanisms directly.

Discrimination and calibration vary together in the native runs. Near-ceiling regimes reach post-scaling ECE near 0.001 (CAMELYON16 patch), whereas the grading and subtyping regimes retain residual post-scaling ECE up to 0.10–0.17. The independently computed probes in RQ3 test whether frozen-feature separability explains this pattern better than ρ or effective sample size; the CE test score itself is not used as explanatory evidence.

5 Discussion

(a) Frozen-encoder bottleneck. Once a strong frozen pathology encoder is fixed, patch-level gains come from reshaping the decision head or objective rather than from resampling or reweighting. CFAL replaces the linear head with learnable class prototypes and a margin objective, pulling examples toward compact class centres and penalising confusion with nearby wrong-class prototypes. OKO constructs training sets with one matched pair and k distractors; averaging per-example logits across set members implicitly regularises logit magnitude and suppresses the over-confidence that afflicts CE-trained classifiers, which explains its combination of high balanced accuracy and the best raw NLL and Brier among patch methods. The simpler reweighting and resampling variants remain within the CE overlap band, consistent with the interpretation that exposure corrections address only part of the error pattern when the representation already separates most classes. ProGAN augmentation does not deliver a clear advantage despite pixel-level FID comparable to the source study (Ruiz-Casado et al., 2024); the decisive difference is that the source method trains its classifier end-to-end on pixels, whereas this benchmark feeds synthetic patches through a frozen encoder that may map slightly off-manifold synthetic patches into embedding regions the head rarely sees for real data. On WSI bags, RankMix and MDE-MIL improve on plain MIL CE in every seed, because the bottleneck there includes instance selection and bag aggregation, which mixing informative subsequences (RankMix) and coupling natural and balanced branches with cross-expert consistency (MDE-MIL) can address even when the underlying features are strong.

(b) Candidate predictors of recovery. The native runs motivate class separability as a candidate predictor but do not establish it as the governing cause. BRACS is mildly imbalanced yet difficult, whereas CAMELYON16 is severely imbalanced at patch level but near ceiling; PANDA likewise presents detection and grading targets of different difficulty on the same biopsies. These contrasts confound imbalance with task, support, and prediction unit, and baseline performance mechanically limits the possible gain over CE. The controlled deficits and recovery fractions in Equation 5, together with the independent feature-space probes and effective-sample-size estimates, provide the non-circular test required to compare these explanations.

(c) ProGAN frozen-feature hypothesis. ProGAN augmentation is informative on BRACS and PANDA: the mild native imbalances never create the tail-scarcity regime where synthetic minority augmentation helped on TCGA-UT at harsher constructed severities. On BRACS (BAcc 0.460 versus CE 0.451) and PANDA (macro F1 0.855 versus CE 0.859), ProGAN lands within between-seed noise of the CE baseline, neither helping nor hurting. This is consistent with the frozen-feature explanation rather than a generator-quality failure: the generators reach reasonable pixel-level FID, but their outputs must pass through a frozen encoder that may map slightly off-manifold synthetic patches into embedding regions the head rarely sees for real data.

(d) Limitations. TCGA-UT patch labels come from pathologist-selected tumor regions; they are useful patch-level class labels but not dense annotations. Within each dataset the two benchmarks differ in prediction unit and per-slide evidence budget, so their result tables are non-transferable. The validation sweep varies one imbalance-specific control per method and does not exhaust architecture or training-schedule choices, and both regimes use frozen features rather than end-to-end encoder adaptation; the benchmark is controlled rather than a full reproduction of each original paper. Calibration results for the BRACS and PANDA WSI-bag regimes are additionally affected by small test sets and fine-grained label sets rather than method-specific weaknesses. Most importantly, the reported native-distribution results lack the size-matched balanced CE condition required for D_M and R_M ; they therefore cannot by themselves answer RQ1–RQ3. Those questions apply to the controlled follow-up defined in §3.1.

6 Conclusion

This report benchmarked representative class-imbalance mitigation methods on four histopathology datasets using frozen Virchow2 features, bounded evidence budgets, and patient-disjoint splits. The completed native-distribution runs show heterogeneous method gains and calibration across patch and WSI supervision, but they do not isolate imbalance-induced damage. The controlled design therefore first measures the balanced-to-imbalanced CE deficit, then expresses mitigation performance as the fraction of that deficit recovered while reporting calibration and computational costs separately. Independent separability probes, cluster-adjusted effective sample size, and head-to-tail ratio are compared as predictors of recovery, with an additional CAMELYON16 analysis separating rare positive slides from sparse informative instances. This framing distinguishes general task difficulty from damage caused specifically by training imbalance and prevents a mechanically low CE baseline from being interpreted as mitigation headroom.

A Training and Tuning Protocol

The shared training protocol (Table 9), per-method imbalance-control tuning grids (Table 10), and ProGAN generator settings (Table 11) referenced in Section 3.

Setting	Patch benchmark	WSI-bag benchmark
Optimizer	AdamW	AdamW
Learning rate	10^{-3}	10^{-3}
Weight decay	10^{-4}	10^{-4}
Epochs	30	30
Early stopping	none	none
Minibatch size	256 patches	32 slide bags
Model selection	final epoch	final epoch
Random seeds	0, 1, 2	0, 1, 2

Table 9: Shared training protocol for both benchmarks. All methods within a benchmark use the same optimizer, schedule, and epoch budget unless noted in Queisler (2026b).

Regime	Method	Reference default	Validation grid
Patch	Weighted CE	Weight power = 1	{0.25, 0.5, 0.75, 1}
WSI bag	Weighted CE	Weight power = 1	{0, 0.125, 0.25, 0.5, 0.75, 1}
Patch, WSI	Balanced sampler	Sampler power = 1	{0.5, 0.75, 1}
Patch, WSI	Focal loss	$\gamma = 2$	$\gamma \in \{0.5, 1, 1.5, 2\}$
Patch	CE + soft F1	Metric-loss weight = 1	{0.25, 0.5, 1, 2, 4, 8, 16, 32, 64, 256, 512, 1024}
Patch	CE + soft MCC	Metric-loss weight = 1	{0.25, 0.5, 1, 2, 8, 32}
Patch	CFAL	$\gamma = 2, \sigma = 1$	$\sigma \in \{0.25, 1, 4\}$
Patch	ProGAN augmentation	Final-depth epochs = 25	{10, 25, 50}
Patch	OKO	$k = 1$	$k \in \{1, 2, 3, 4, 5, 6, 8, 12, 20\}$
WSI bag	RankMix	$\alpha = 1$	$\alpha \in \{0.2, 0.5, 1, 2, 4, 8, 16, 32, 128\}$
WSI bag	SC-MIL	$\tau = 1$	$\tau \in \{0.05, 0.1, 0.2, 0.5\}$
WSI bag	MDE-MIL	$\lambda_{\text{con}} = 0.25$	$\lambda_{\text{con}} \in \{0.1, 0.25, 0.3, 0.5, 1, 2, 4\}$

Table 10: Method-specific imbalance controls. Each row varies one control on validation while retaining the shared budget in Table 9. Identical grids are applied on all four datasets.

B Controlled-Imbalance Target Distributions

Full native-versus-target class distributions for the controlled construction of §3.1, at both patch and slide level for all four datasets (Figure 5). These are the per-class detail behind the summary ratios of Figure 3 and Table 3.

C Additional TCGA-UT Analysis

This appendix collects the deeper TCGA-UT diagnostics referenced in Section 4.2: validation-selection provenance and tuning sensitivity, paired seed-level comparisons, classwise difficulty against support, and ProGAN augmentation diagnostics.

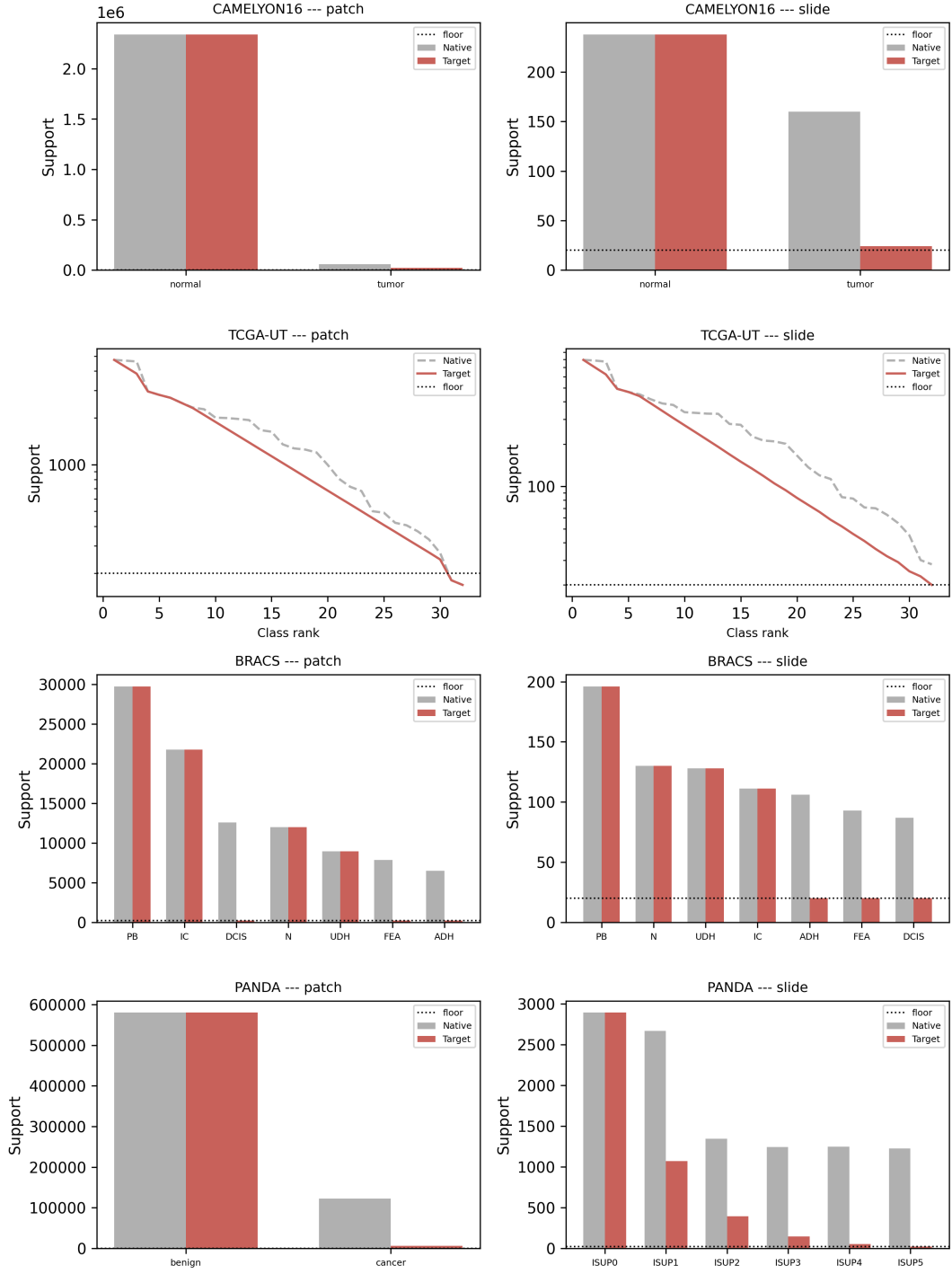


Figure 5: Native (grey) versus proposed controlled-imbalance target (red) support per class, at patch (left) and slide (right) level, one row per dataset in the order of Table 1, with the floor N_{floor} (dotted). CAMELYON16 and PANDA-patch use binary ratio downsampling; TCGA-UT uses truncated exponential decay (rank plots, log scale); BRACS uses step imbalance; PANDA-slide uses ordinal exponential decay.

ProGAN synthetic patches and nearest real training neighbors (seed 0, Virchow2 embedding space)

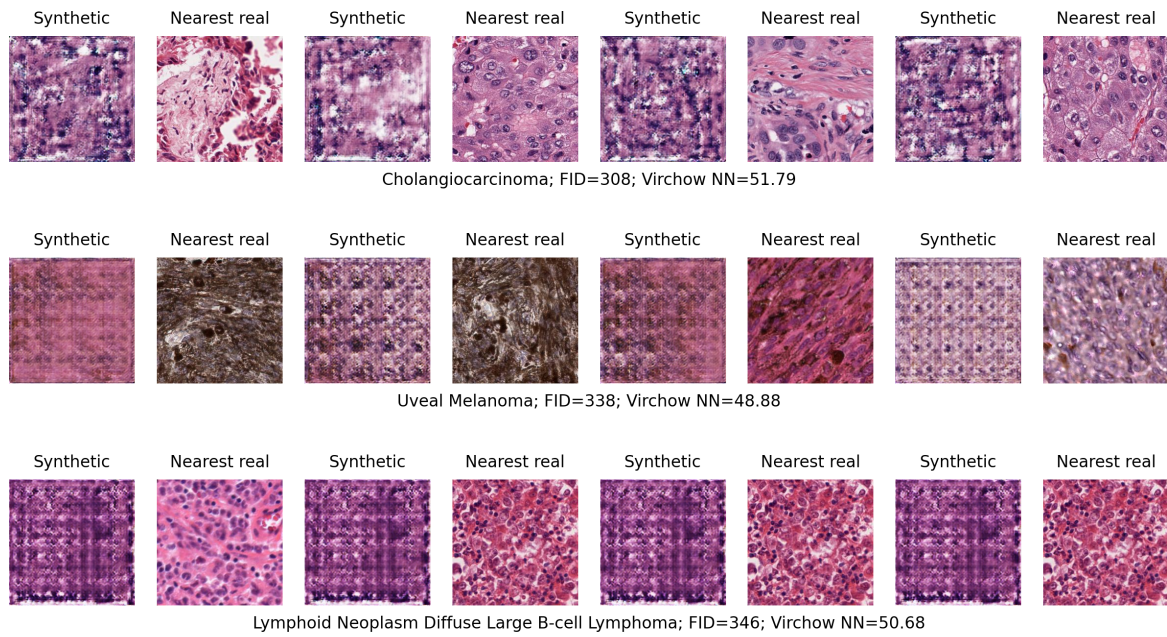


Figure 6: ProGAN synthetic patches (left column of each pair) and their nearest real training neighbors in Virchow2 embedding space (right column) for three tail classes in seed 0. The frozen-feature pipeline—not generator quality—is the proposed explanation for the null downstream result.

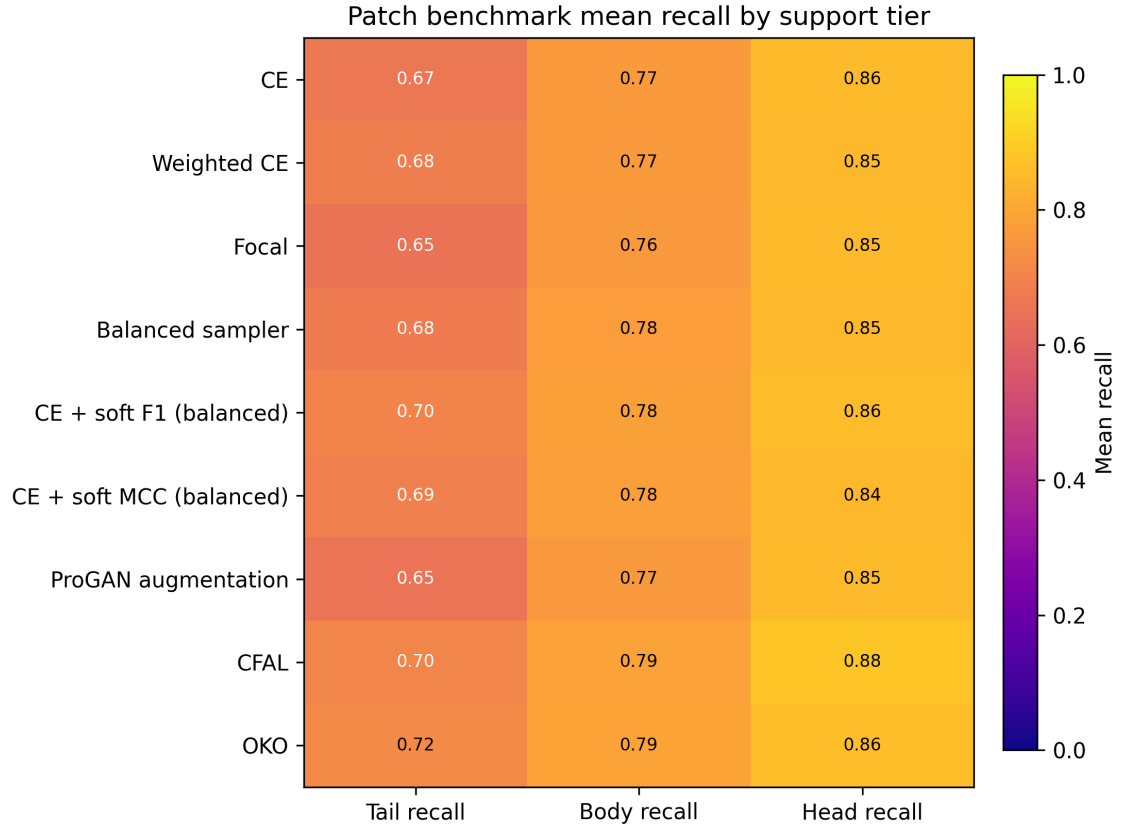


Figure 7: TCGA-UT patch-level mean recall by support tier on the held-out test split, exposing the head–body–tail variation hidden by aggregate accuracy.

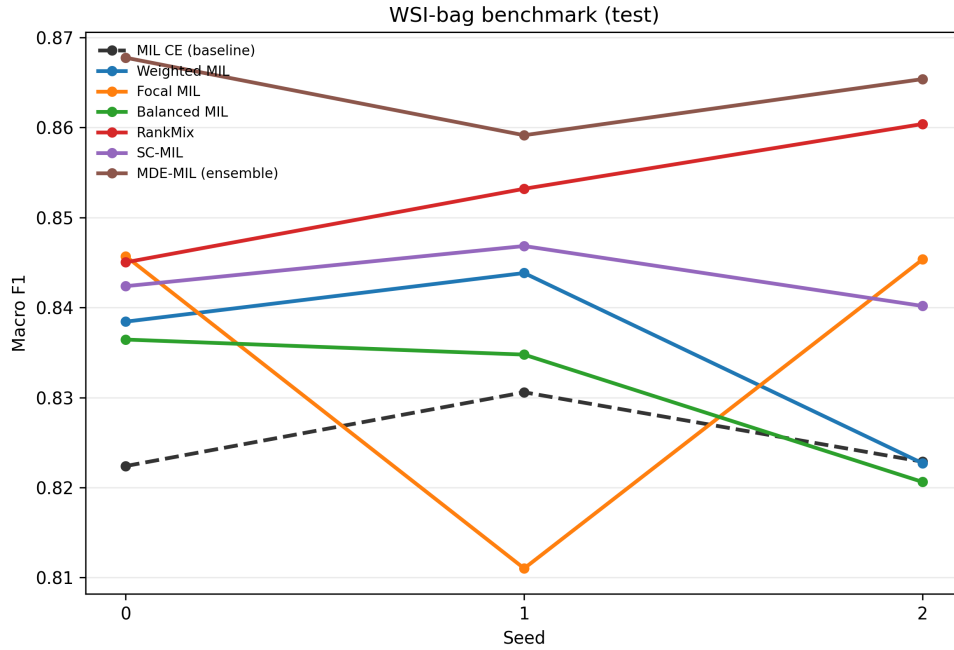


Figure 8: Validation-selected TCGA-UT WSI-bag macro F1 on the held-out test split by random seed. The dashed line marks plain MIL CE; the pronounced focal-MIL dip on seed 1 is a single-seed training instability.

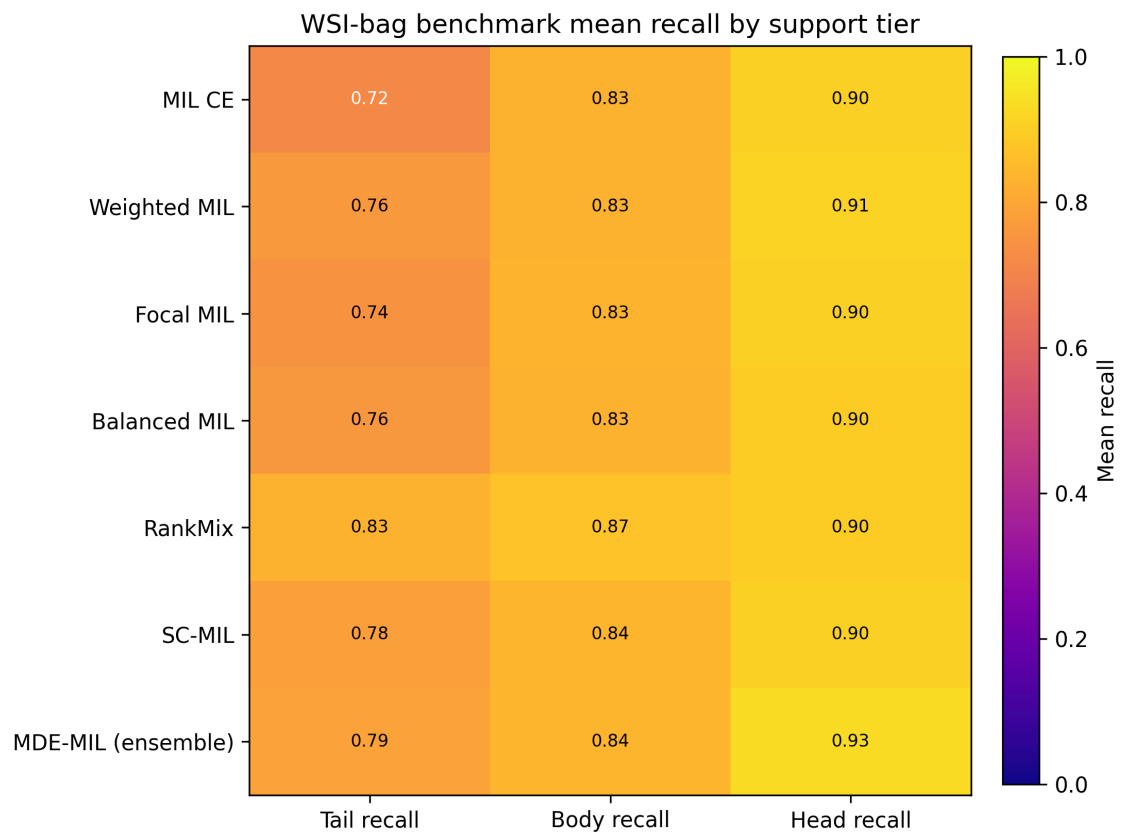


Figure 9: TCGA-UT WSI-bag classwise recall on the held-out test split. The spread across classes shows why aggregate accuracy alone is insufficient in the imbalanced setting.

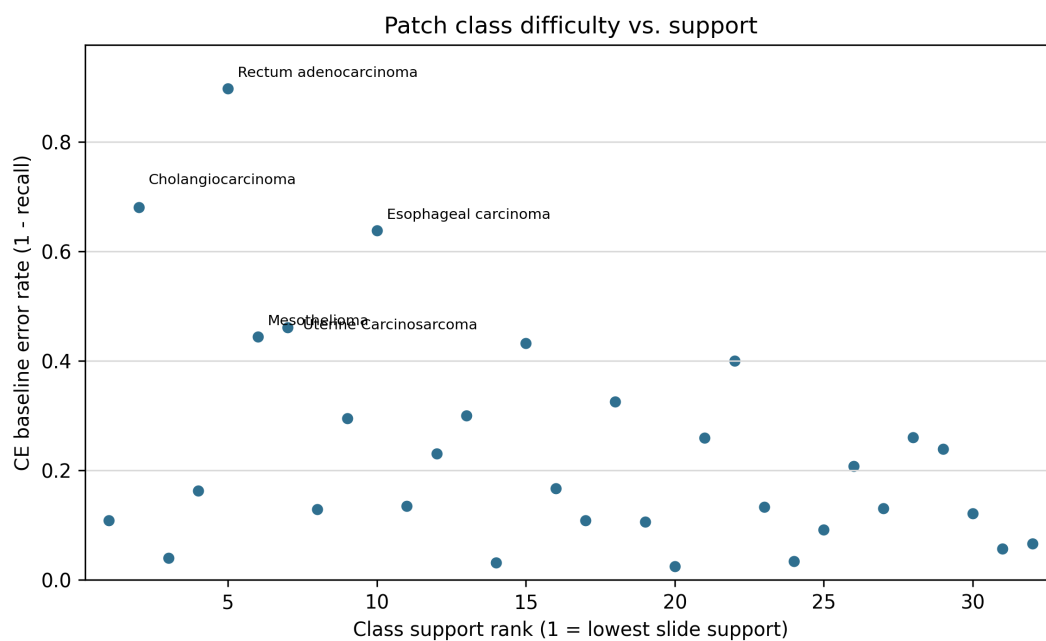


Figure 10: TCGA-UT patch CE-baseline class error rate versus full-dataset class-support rank. The hardest patch classes are not simply ordered by rarity.

Setting	Value
Resolution stages	4×4 to 256×256 (seven stages)
Epochs per stage (depths 1–6)	25
Final-depth (256×256) refinement epochs	25 (reference); validation selects from $\{10, 25, 50\}$
Fade-in fraction for new layers	0.5 (pinned to the reference 25-epoch schedule)
Latent dimension	128
Base channel width	512
Adversarial loss	WGAN-GP ($\lambda = 10$, drift = 10^{-3})
Equalized learning rate	Yes (weight $\sim \mathcal{N}(0, 1)$, scale $\sqrt{2/\text{fan_in}}$ at forward)
Optimizer	Adam, learning rate 2×10^{-4} , $\beta_1 = 0.5$
Real patches per class for GAN training	Up to 2,048, deterministic per-class subsample
Synthetic count per tail class	Head-class train count minus class train count

Table 11: ProGAN generator training settings (TCGA-UT patch regime only). Depths 1–6 use 25 epochs each; for the final depth one long run trains to 50 epochs, snapshotting the generator at epochs 10, 25, and 50, and validation selects the snapshot with the best mean macro F1 over seeds, reusing one GAN training run for all three sweep points.

D Native-Dataset Calibration Tables

Raw and temperature-scaled (TS) calibration for CAMELYON16, BRACS, and PANDA, referenced in Section 4. Each table gives ECE before and after scaling and the fitted temperature T , mean $\pm$ standard deviation across seeds.

Regime	Method	Selected control	Val macro F1	Test macro F1	Test bal. acc.
Patch	CE	baseline (no sweep)	–	0.773	0.764
Patch	Balanced sampler	sampler_power=1	0.768	0.771	0.769
Patch	CE + soft F1 (balanced)	metric_loss_weight=256	0.781	0.780	0.781
Patch	CE + soft MCC (balanced)	metric_loss_weight=32	0.771	0.770	0.772
Patch	CFAL	cfal_sigma=1	0.792	0.798	0.789
Patch	Focal	focal_gamma=1.5	0.764	0.765	0.758
Patch	OKO	oko_k=4	0.783	0.788	0.791
Patch	ProGAN augmentation	final_depth_epochs=10	0.756	0.766	0.760
Patch	Weighted CE	weight_power=0.5	0.764	0.774	0.767
WSI bag	MIL CE	baseline (no sweep)	–	0.826	0.823
WSI bag	MDE-MIL (ensemble)	mde_mil_consistency_weight=2	0.851	0.865	0.849
WSI bag	Balanced MIL	sampler_power=0.75	0.821	0.831	0.829
WSI bag	Focal MIL	focal_gamma=0.5	0.823	0.824	0.823
WSI bag	Weighted MIL	weight_power=0.125	0.828	0.835	0.833
WSI bag	RankMix	rankmix_alpha=32	0.843	0.854	0.864
WSI bag	SC-MIL	sc_mil_temperature=0.1	0.828	0.839	0.837

Table 12: Validation-selected TCGA-UT configuration provenance. Configurations are chosen by mean validation macro F1 over seeds 0–2 with balanced accuracy as tie-breaker. A seed-blocked repeated-measures analysis of the validation sweeps shows across-configuration variation exceeding twice the seed standard deviation only for OKO (range-to-seed-noise ratio 75.8) and RankMix (ratio 2.06, $p < 0.01$); for all single-knob resampling, reweighting, and focal mechanisms the swept control produces variation within seed noise regardless of its value.

Comparison	Δ Macro F1	Δ Balanced accuracy
CE + soft MCC – CE	-0.003 ± 0.008 [+0.005, –0.000, –0.014]	$+0.009 \pm 0.007$ [+0.015, +0.012, –0.000]
CE + soft F1 – CE	$+0.007 \pm 0.010$ [+0.014, +0.014, –0.007]	$+0.018 \pm 0.008$ [+0.023, +0.023, +0.007]
OKO – CE	$+0.015 \pm 0.010$ [+0.024, +0.021, +0.001]	$+0.028 \pm 0.012$ [+0.038, +0.033, +0.011]
CFAL – CE	$+0.025 \pm 0.005$ [+0.030, +0.027, +0.019]	$+0.026 \pm 0.004$ [+0.030, +0.027, +0.020]
RankMix – MIL CE	$+0.028 \pm 0.010$ [+0.023, +0.041, +0.020]	$+0.041 \pm 0.010$ [+0.038, +0.055, +0.031]
MDE-MIL – MIL CE	$+0.039 \pm 0.016$ [+0.055, +0.043, +0.018]	$+0.026 \pm 0.016$ [+0.041, +0.034, +0.003]

Table 13: TCGA-UT paired validation-selected test-split differences against the within-regime CE baseline on seeds 0–2. Each cell reports mean $\pm$ SD across seeds; bracketed values list seed-ordered paired deltas. CFAL and OKO are above plain CE on macro F1 and balanced accuracy in every seed; on WSI bags, both RankMix and MDE-MIL are above plain MIL CE on both headline metrics in every seed. No formal p -values are computed.

Regime	Method	NLL	Brier	ECE
Patch	CFAL	2.850 ± 0.011	0.914 ± 0.001	0.787 ± 0.008
Patch	OKO	0.756 ± 0.040	0.270 ± 0.015	0.158 ± 0.013
Patch	CE + soft F1 (balanced)	3.038 ± 0.137	0.316 ± 0.008	0.155 ± 0.004
Patch	Weighted CE	1.965 ± 0.115	0.314 ± 0.016	0.147 ± 0.007
Patch	CE	1.976 ± 0.149	0.312 ± 0.018	0.147 ± 0.009
Patch	Balanced sampler	1.967 ± 0.139	0.313 ± 0.019	0.147 ± 0.009
Patch	CE + soft MCC (balanced)	2.870 ± 0.153	0.332 ± 0.017	0.162 ± 0.008
Patch	ProGAN augmentation	2.511 ± 0.093	0.327 ± 0.007	0.157 ± 0.004
Patch	Focal	1.370 ± 0.081	0.298 ± 0.014	0.131 ± 0.006
WSI bag	MDE-MIL (ensemble)	0.487 ± 0.005	0.166 ± 0.002	0.036 ± 0.004
WSI bag	RankMix	0.584 ± 0.029	0.170 ± 0.006	0.049 ± 0.001
WSI bag	SC-MIL	1.024 ± 0.145	0.204 ± 0.020	0.092 ± 0.009
WSI bag	Weighted MIL	1.224 ± 0.144	0.212 ± 0.012	0.099 ± 0.006
WSI bag	Balanced MIL	1.264 ± 0.165	0.223 ± 0.010	0.102 ± 0.005
WSI bag	MIL CE	1.283 ± 0.080	0.227 ± 0.024	0.106 ± 0.012
WSI bag	Focal MIL	1.117 ± 0.153	0.221 ± 0.014	0.098 ± 0.009

Table 14: TCGA-UT held-out test raw calibration metrics for validation-selected configurations over three seeds. Lower is better for NLL, Brier, and ECE. Values are not comparable across regimes. Table 15 gives the post-hoc counterpart.

Regime	Method	NLL	Brier	ECE
Patch	CFAL	0.532 ± 0.029	0.220 ± 0.011	0.019 ± 0.003
Patch	OKO	0.617 ± 0.032	0.236 ± 0.011	0.029 ± 0.006
Patch	CE + soft F1 (balanced)	0.780 ± 0.026	0.263 ± 0.008	0.061 ± 0.028
Patch	Weighted CE	0.654 ± 0.030	0.252 ± 0.012	0.038 ± 0.005
Patch	CE	0.649 ± 0.035	0.250 ± 0.014	0.041 ± 0.002
Patch	Balanced sampler	0.653 ± 0.034	0.252 ± 0.013	0.038 ± 0.008
Patch	CE + soft MCC (balanced)	0.768 ± 0.034	0.271 ± 0.013	0.076 ± 0.029
Patch	ProGAN augmentation	0.723 ± 0.017	0.263 ± 0.005	0.073 ± 0.007
Patch	Focal	0.618 ± 0.025	0.251 ± 0.009	0.006 ± 0.004
WSI bag	MDE-MIL (ensemble)	0.471 ± 0.009	0.163 ± 0.003	0.023 ± 0.002
WSI bag	RankMix	0.428 ± 0.012	0.166 ± 0.005	0.028 ± 0.002
WSI bag	SC-MIL	0.440 ± 0.045	0.173 ± 0.018	0.007 ± 0.004
WSI bag	Weighted MIL	0.455 ± 0.033	0.177 ± 0.011	0.013 ± 0.004
WSI bag	Balanced MIL	0.479 ± 0.031	0.185 ± 0.010	0.013 ± 0.003
WSI bag	MIL CE	0.473 ± 0.018	0.188 ± 0.015	0.017 ± 0.010
WSI bag	Focal MIL	0.475 ± 0.024	0.189 ± 0.008	0.012 ± 0.002

Table 15: TCGA-UT held-out test calibration metrics after method-agnostic post-hoc temperature scaling. For each method and seed the temperature is fit only on that seed’s validation split and applied to the test split. Table 14 gives the raw-score diagnostic.

Regime	Class	Slides	Support rank	Difficulty rank	Recall	F1
Patch	Rectum adenocarcinoma	63	5	1	0.103	0.144
Patch	Cholangiocarcinoma	30	2	2	0.319	0.409
Patch	Esophageal carcinoma	113	10	3	0.362	0.420
Patch	Uterine Carcinosarcoma	71	7	4	0.539	0.519
Patch	Mesothelioma	70	6	5	0.556	0.628
WSI-bag	Rectum adenocarcinoma	63	5	1	0.338	0.409
WSI-bag	Esophageal carcinoma	113	10	2	0.523	0.564
WSI-bag	Cholangiocarcinoma	30	2	3	0.528	0.544
WSI-bag	Uterine Carcinosarcoma	71	7	4	0.625	0.611
WSI-bag	Stomach adenocarcinoma	274	18	5	0.693	0.735

Table 16: Hardest TCGA-UT classes under each regime’s CE baseline, averaged over seeds 0–2 on the held-out test split. Support rank is computed from full TCGA-UT slide counts (rank 1 = lowest support); difficulty rank orders classes by baseline error rate, $1 - \text{recall}$. Rectum adenocarcinoma is the hardest class in both regimes despite being the fifth-lowest-support class, and the rarest class (diffuse large B-cell lymphoma) is not among the five hardest; the rank association between slide support and CE-baseline recall is positive but modest (Spearman $\rho = 0.334$ for patches, $\rho = 0.301$ for WSI bags).

Class	Real train	Generated	Inception FID	Virchow2 mean NN
Lymphoid Neoplasm Diffuse Large B-cell Lymphoma	600	16,240	345.6	50.68
Cholangiocarcinoma	660	16,180	307.6	51.79
Pheochromocytoma and Paraganglioma	900	15,940	382.0	54.06
Uveal Melanoma	1,160	15,680	338.4	48.88
Rectum adenocarcinoma	1,350	15,490	239.5	47.97
Mesothelioma	1,430	15,410	286.3	51.23
Uterine Carcinosarcoma	1,550	15,290	304.0	50.94
Kidney Chromophobe	1,740	15,100	282.8	48.97
All augmented classes (31 total)	165,470	356,570	293.4 median	51.32 median

Table 17: ProGAN augmentation diagnostics for seed 0. “Generated” counts are synthetic patches added per class to reach the head-class training patch count. Inception FID compares generated patches to the real training subset; the class-specific generators reach FID 239.5–382.0 (median 293.4), comparable to the ProGAN FID reported by Ruiz-Casado et al. (2024). Virchow2 mean NN is the mean minimum L_2 distance from a fixed synthetic subsample to real same-class embeddings in the frozen feature cache.

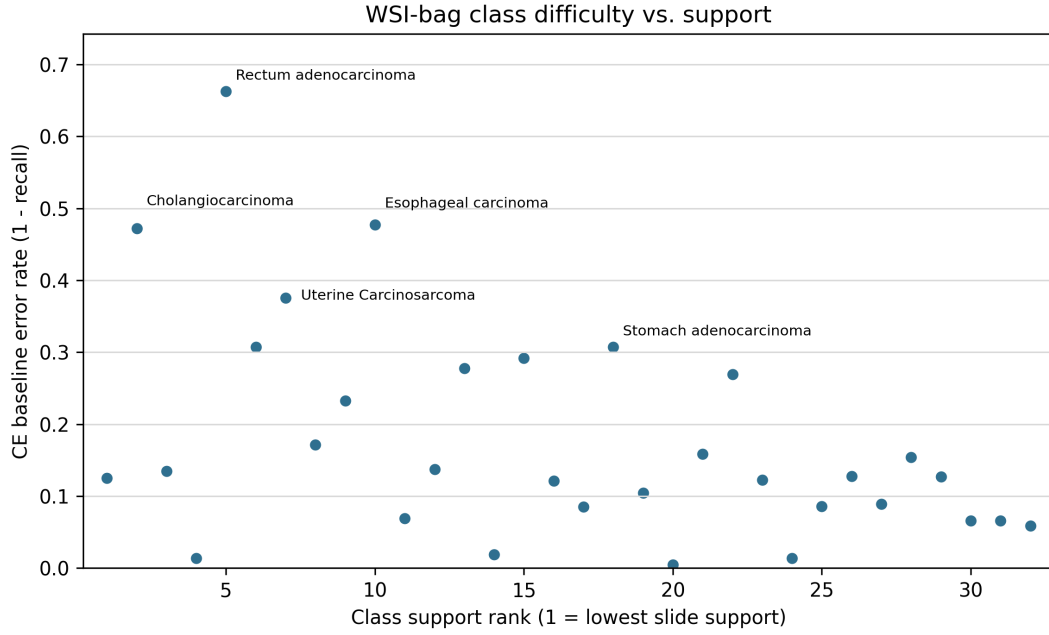


Figure 11: TCGA-UT WSI-bag CE-baseline class error rate versus full-dataset class-support rank. The same high-error classes are visible at slide level, showing that class difficulty is not reducible to class frequency.

Method	ECE	ECE+TS	T
CE	0.003 ± 0.001	0.001 ± 0.001	3.723 ± 0.142
Weighted CE	0.004 ± 0.002	0.001 ± 0.001	4.224 ± 0.503
Focal loss	0.003 ± 0.001	0.001 ± 0.000	4.547 ± 0.173
CFAL	0.464 ± 0.001	0.068 ± 0.004	0.050 ± 0.000
CE + soft F1	0.004 ± 0.001	0.001 ± 0.001	4.598 ± 0.179
CE + soft MCC	0.005 ± 0.001	0.002 ± 0.000	5.025 ± 0.191
Balanced sampling	0.003 ± 0.001	0.001 ± 0.001	4.071 ± 0.239
OKO	0.003 ± 0.000	0.001 ± 0.000	3.014 ± 0.100
ProGAN augmentation	0.003 ± 0.001	0.001 ± 0.001	4.119 ± 0.281

Table 18: CAMELYON16 native patch-feature calibration.

Method	ECE	ECE+TS	T
MIL CE	0.027 ± 0.017	0.034 ± 0.017	0.050 ± 0.000
Weighted CE	0.027 ± 0.017	0.034 ± 0.017	0.050 ± 0.000
Focal loss	0.021 ± 0.013	0.023 ± 0.010	0.050 ± 0.000
Balanced sampling	0.022 ± 0.009	0.023 ± 0.010	0.050 ± 0.000
RankMix	0.022 ± 0.030	0.028 ± 0.024	0.273 ± 0.386
SC-MIL	0.020 ± 0.018	0.017 ± 0.017	0.050 ± 0.000
MDE-MIL	0.013 ± 0.008	0.017 ± 0.017	0.050 ± 0.000

Table 19: CAMELYON16 native WSI-bag calibration.

Method	ECE	ECE+TS	T
CE	0.354 ± 0.039	0.031 ± 0.010	5.812 ± 0.409
Weighted CE	0.367 ± 0.039	0.038 ± 0.017	5.752 ± 0.486
Focal loss	0.286 ± 0.023	0.030 ± 0.017	3.459 ± 0.405
CFAL	0.417 ± 0.026	0.049 ± 0.013	0.079 ± 0.010
CE + soft F1	0.405 ± 0.030	0.070 ± 0.027	8.532 ± 1.329
CE + soft MCC	0.366 ± 0.039	0.029 ± 0.011	6.582 ± 0.723
Balanced sampling	0.347 ± 0.036	0.021 ± 0.011	5.823 ± 0.601
OKO	0.045 ± 0.028	0.039 ± 0.014	0.877 ± 0.149
ProGAN augmentation	0.358 ± 0.041	0.039 ± 0.018	6.158 ± 0.677

Table 20: BRACS native patch-feature calibration.

Method	ECE	ECE+TS	T
MIL CE	0.237 ± 0.012	0.165 ± 0.040	3.092 ± 0.322
Weighted CE	0.247 ± 0.051	0.163 ± 0.045	2.928 ± 0.343
Focal loss	0.164 ± 0.066	0.136 ± 0.010	1.977 ± 0.075
Balanced sampling	0.257 ± 0.064	0.168 ± 0.052	2.861 ± 0.314
RankMix	0.151 ± 0.074	0.144 ± 0.017	1.298 ± 0.108
SC-MIL	0.302 ± 0.097	0.168 ± 0.014	2.421 ± 0.242
MDE-MIL	0.277 ± 0.017	0.151 ± 0.010	2.083 ± 0.326

Table 21: BRACS native WSI-bag calibration.

Method	ECE	ECE+TS	T
CE	0.046 ± 0.007	0.035 ± 0.003	2.733 ± 0.236
Weighted CE	0.045 ± 0.005	0.036 ± 0.002	2.703 ± 0.225
Focal loss	0.024 ± 0.000	0.031 ± 0.002	1.404 ± 0.126
CFAL	0.382 ± 0.004	0.011 ± 0.003	0.060 ± 0.001
CE + soft F1	0.103 ± 0.009	0.009 ± 0.001	6.418 ± 0.324
CE + soft MCC	0.101 ± 0.008	0.008 ± 0.003	5.491 ± 0.183
Balanced sampling	0.072 ± 0.005	0.019 ± 0.003	3.642 ± 0.184
OKO	0.028 ± 0.008	0.005 ± 0.002	0.822 ± 0.027
ProGAN augmentation	0.042 ± 0.007	0.039 ± 0.002	2.595 ± 0.343

Table 22: PANDA native patch-feature calibration.

Method	ECE	ECE+TS	T
MIL CE	0.237 ± 0.024	0.058 ± 0.013	3.410 ± 0.240
Weighted CE	0.237 ± 0.024	0.058 ± 0.013	3.410 ± 0.240
Focal loss	0.129 ± 0.022	0.062 ± 0.020	2.002 ± 0.166
Balanced sampling	0.241 ± 0.021	0.039 ± 0.009	3.723 ± 0.142
RankMix	0.144 ± 0.009	0.045 ± 0.020	1.530 ± 0.029
SC-MIL	0.252 ± 0.020	0.070 ± 0.015	3.452 ± 0.297
MDE-MIL	0.239 ± 0.035	0.097 ± 0.039	2.113 ± 0.080

Table 23: PANDA native WSI-bag calibration.

E Terminology

Term	Meaning in this study
<i>Slide / WSI</i>	Diagnostic whole-slide image carrying one label.
<i>Case / patient</i>	TCGA participant identified by the first three fields of the TCGA slide barcode (e.g. TCGA-OR-A5J1); or the equivalent patient identifier in BRACS and PANDA.
<i>Patch</i>	256×256 H&E crop from a selected region; patch labels inherit the slide-level class.
<i>Patch embedding</i>	Frozen Virchow2 vector for one controlled patch; input to the patch-level classifier head.
<i>WSI bag</i>	Set of patch-feature instances for one slide, consumed by the MIL model.
<i>Imbalance factor ρ</i>	Head-to-tail ratio $N_{\max}/N_{\min} \geq 1$ (the long-tailed-recognition imbalance factor); primary imbalance measure, reported at patch and slide level.
<i>Imbalance $1 - H_{\text{norm}}$</i>	Entropy-based class-imbalance measure in $[0, 1]$ (0 balanced, 1 maximally skewed); class-count-invariant cross-check to ρ , reported at patch and slide level (Queisler, 2026a).
<i>Head/body/tail class</i>	Fixed support tier from the full TCGA-UT slide distribution (8 each for head and tail; 16 body).
<i>Case-disjoint split</i>	Split rule preventing any participant’s slides, patches, or features from crossing train, validation, and test.
<i>Evidence budget</i>	Per-slide cap on the number of patch or bag instances; dataset-specific, following a common data-driven median rule (Table 1).

Table 24: Terminology used across all four dataset benchmarks.

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